

Section Properties Document Page 2 below contains the section properties for the Non-Structural Studs.

NR-25-03-17 NON-STRUCTURAL STUD SECTION PROPERTIES^{1,2,3,4,5}

MEMBER	Design Thickness (in)	F _y (ksi)	GROSS PROPERTIES							EFFECTIVE PROPERTIES					TORSIONAL PROPERTIES				
			Area (in ²)	Weight (lb/ft)	I _x (in ⁴)	S _x (in ³)	r _x (in.)	I _y (in ⁴)	r _y (in.)	I _{xe} (in ⁴)	S _{xe} (in ³)	M _{al} (in-k)	M _{ad} (in-k)	V _{ay} (kip)	Jx1000 (in ⁴)	C _w (in ⁶)	X _o (in.)	r _o (in.)	β
162S125-18	0.0188	33	0.0802	0.2732	0.0378	0.0465	0.686	0.0160	0.447	0.0308	0.0333	0.659	0.653	0.302	0.0094	0.0092	-1.029	1.315	0.3880
162S125-27	0.0283	33	0.1196	0.4076	0.0557	0.0685	0.682	0.0234	0.443	0.0490	0.0554	1.095	1.137	0.494	0.0319	0.0132	-1.017	1.302	0.3899
162S125-30	0.0312	33	0.1315	0.4481	0.0610	0.0750	0.681	0.0256	0.441	0.0548	0.0628	1.242	1.294	0.543	0.0427	0.0144	-1.014	1.299	0.3905
250S125-18	0.0188	33	0.0966	0.3293	0.0993	0.0795	1.014	0.0186	0.439	0.0827	0.0595	1.176	1.028	0.258	0.0114	0.0233	-0.904	1.427	0.5992
250S125-27	0.0283	33	0.1444	0.4920	0.1471	0.1177	1.009	0.0272	0.434	0.1310	0.0976	1.928	1.824	0.685	0.0385	0.0338	-0.893	1.416	0.6025
250S125-30	0.0312	33	0.1588	0.5411	0.1614	0.1291	1.008	0.0298	0.433	0.1464	0.1102	2.178	2.088	0.833	0.0515	0.0369	-0.889	1.412	0.6035
350S125-18	0.0188	33	0.1154	0.3933	0.2154	0.1231	1.366	0.0207	0.423	0.1738	0.0873	1.725	1.468	0.180	0.0136	0.0498	-0.797	1.637	0.7628
350S125-27	0.0283	33	0.1727	0.5884	0.3198	0.1828	1.361	0.0302	0.418	0.2879	0.1550	3.062	2.649	0.614	0.0461	0.0725	-0.787	1.627	0.7659
350S125-30	0.0312	33	0.1900	0.6475	0.3512	0.2007	1.360	0.0331	0.417	0.3213	0.1745	3.448	3.046	0.824	0.0616	0.0791	-0.784	1.624	0.7669
362S125-18	0.0188	33	0.1178	0.4013	0.2337	0.1290	1.409	0.0209	0.421	0.1872	0.0903	1.784	1.522	0.173	0.0139	0.0539	-0.786	1.667	0.7777
362S125-27	0.0283	33	0.1762	0.6005	0.3472	0.1916	1.404	0.0305	0.416	0.3129	0.1628	3.217	2.753	0.592	0.0470	0.0785	-0.776	1.657	0.7808
362S125-30	0.0312	33	0.1939	0.6608	0.3813	0.2104	1.402	0.0334	0.415	0.3492	0.1833	3.621	3.167	0.793	0.0629	0.0857	-0.773	1.654	0.7817
400S125-18 ⁶	0.0188	33	0.1248	0.4254	0.2944	0.1472	1.536	0.0214	0.415	0.2309	0.0994	1.963	1.685	0.156	0.0147	0.0675	-0.754	1.761	0.8165
400S125-27	0.0283	33	0.1868	0.6366	0.4377	0.2189	1.531	0.0314	0.410	0.3959	0.1873	3.701	3.065	0.533	0.0499	0.0984	-0.744	1.751	0.8193
400S125-30	0.0312	33	0.2056	0.7006	0.4808	0.2404	1.529	0.0343	0.409	0.4416	0.2106	4.161	3.532	0.715	0.0667	0.1074	-0.741	1.748	0.8202
550S125-18 ⁷	0.0188	33	0.1530	0.5215	0.6300	0.2291	2.029	0.0232	0.390	0.4514	0.1322	2.612	2.293	0.112	0.0180	0.1404	-0.651	2.166	0.9098
550S125-27	0.0283	33	0.2293	0.7813	0.9386	0.3413	2.023	0.0340	0.385	0.7756	0.2463	4.866	4.266	0.382	0.0612	0.2050	-0.641	2.157	0.9116
550S125-30	0.0312	33	0.2524	0.8601	1.0315	0.3751	2.022	0.0372	0.384	0.8817	0.2860	5.652	4.947	0.512	0.0819	0.2238	-0.639	2.155	0.9121
600S125-18 ⁸	0.0188	33	0.1624	0.5535	0.7786	0.2595	2.189	0.0237	0.382	--	--	--	--	--	0.0191	0.1715	-0.623	2.308	0.9272
600S125-27 ⁶	0.0283	33	0.2434	0.8295	1.1605	0.3868	2.184	0.0347	0.378	0.9420	0.2708	5.351	4.639	0.349	0.0650	0.2505	-0.614	2.299	0.9288
600S125-30	0.0312	33	0.2680	0.9133	1.2756	0.4252	2.182	0.0379	0.376	1.0723	0.3149	6.223	5.390	0.467	0.0870	0.2736	-0.611	2.297	0.9292

¹Properties are based on the full, unreduced cross-section away from any web punch-outs.

²Use the effective moment of inertia for deflection calculation.

³Allowable moment is lesser of M_{al} and M_{ad}. Distortional buckling is based on an assumed KΦ = 0.

⁴The effect of cold-work of forming on strength was conservatively not incorporated.

⁵See technical manual for product identification information. Calculations were performed using dimensions specified in the client's technical manual dated 7/29/2025.

⁶Web height-to-thickness ratio, h/t, exceeds 200 but is less than 260. Bearing stiffeners are required in accordance with AISI S100. No holes/punchouts are permitted in the web at these locations.

⁷Web height-to-thickness ratio, h/t, exceeds 260 but is less than 300. Bearing and intermediate stiffeners must be provided in accordance with AISI S100.

⁸Web height-to-thickness ratio, h/t, exceeds 300. Allowable design values for these members are outside the scope and may be determined in accordance with AISI S100.

SYMBOLS

I_x = Moment of inertia about x-axis
S_x = Section modulus about x-axis
r_x = Radius of gyration about x-axis
I_y = Moment of inertia about y-axis
r_y = Radius of gyration about y-axis

M_{al} = Allowable bending moment about x-axis based on local buckling
M_{ad} = Allowable moment about x-axis based on distortional buckling
V_{ay} = Allowable shear

J = St. Venant torsion constant
C_w = Torsional warping constant
X_o = Distance from shear center to the centroid along the principal X-axis
r_o = Polar radius of gyration about shear center
β = Torsional flexural constant

Section Properties Document Pages 4-6 below contain the section properties for the Structural Studs.

NR-25-03-17 STRUCTURAL STUD SECTION PROPERTIES^{1,2,3,4,5}

MEMBER	Design Thickness (in)	F _y (ksi)	GROSS PROPERTIES							EFFECTIVE PROPERTIES					TORSIONAL PROPERTIES				
			Area (in ²)	Weight (lb/ft)	I _x (in ⁴)	S _x (in ³)	r _x (in.)	I _y (in ⁴)	r _y (in.)	I _{xe} (in ⁴)	S _{xe} (in ³)	M _{al} (in-k)	M _{ad} (in-k)	V _{ay} (kip)	Jx1000 (in ⁴)	C _w (in ⁶)	X _o (in.)	r _o (in.)	β
250S137-33	0.0346	33	0.1972	0.6722	0.2033	0.1626	1.015	0.0524	0.515	0.1994	0.1571	3.105	3.095	0.975	0.0787	0.0764	-1.141	1.612	0.4987
250S137-43	0.0451	33	0.2552	0.8697	0.2606	0.2085	1.011	0.0665	0.511	0.2609	0.2089	4.128	4.120	1.265	0.1730	0.0959	-1.129	1.599	0.5013
250S137-54	0.0566	50	0.3158	1.0762	0.3182	0.2546	1.004	0.0802	0.504	0.3132	0.2475	7.409	7.622	2.353	0.3372	0.1154	-1.115	1.583	0.5037
250S137-68	0.0713	50	0.3900	1.3292	0.3859	0.3087	0.995	0.0956	0.495	0.3870	0.3103	9.290	9.243	2.866	0.6609	0.1378	-1.096	1.561	0.5068
250S137-97	0.1017	50	0.5333	1.8176	0.5071	0.4056	0.975	0.1207	0.476	0.5106	0.4105	12.290	12.145	3.798	1.8387	0.1757	-1.057	1.515	0.5130
250S162-33	0.0346	33	0.2145	0.7312	0.2296	0.1836	1.034	0.0783	0.604	0.2120	0.1604	3.170	3.268	0.975	0.0856	0.1131	-1.373	1.822	0.4322
250S162-43	0.0451	33	0.2778	0.9466	0.2946	0.2357	1.030	0.0999	0.600	0.2870	0.2248	4.443	4.574	1.265	0.1883	0.1424	-1.361	1.809	0.4340
250S162-54	0.0566	50	0.3441	1.1726	0.3605	0.2884	1.024	0.1210	0.593	0.3478	0.2705	8.100	8.127	2.353	0.3674	0.1720	-1.346	1.792	0.4357
250S162-68	0.0713	50	0.4257	1.4507	0.4385	0.3508	1.015	0.1451	0.584	0.4301	0.3386	10.137	10.502	2.866	0.7213	0.2064	-1.327	1.770	0.4377
250S162-97	0.1017	50	0.5842	1.9909	0.5802	0.4641	0.997	0.1859	0.564	0.5837	0.4691	14.045	13.897	3.798	2.0140	0.2656	-1.287	1.723	0.4418
250S200-33	0.0346	33	0.2405	0.8196	0.2690	0.2152	1.058	0.1298	0.735	0.2321	0.1681	3.322	3.466	0.975	0.0960	0.1854	-1.726	2.153	0.3577
250S200-43	0.0451	33	0.3116	1.0619	0.3456	0.2764	1.053	0.1661	0.730	0.3119	0.2321	4.586	4.906	1.265	0.2113	0.2342	-1.713	2.139	0.3588
250S200-54	0.0566	50	0.3865	1.3173	0.4238	0.3391	1.047	0.2021	0.723	0.3769	0.2781	8.325	8.669	2.353	0.4128	0.2841	-1.698	2.122	0.3596
250S200-68	0.0713	50	0.4791	1.6329	0.5173	0.4139	1.039	0.2441	0.714	0.4903	0.3767	11.278	11.543	2.866	0.8119	0.3424	-1.679	2.099	0.3606
250S200-97	0.1017	50	0.6604	2.2508	0.6899	0.5519	1.022	0.3177	0.694	0.6839	0.5426	16.245	16.524	3.798	2.2770	0.4454	-1.638	2.052	0.3624
350S137-33	0.0346	33	0.2318	0.7901	0.4416	0.2523	1.380	0.0587	0.503	0.4328	0.2439	4.819	4.546	1.024	0.0925	0.1528	-1.016	1.786	0.6763
350S137-43	0.0451	33	0.3003	1.0234	0.5678	0.3245	1.375	0.0746	0.499	0.5681	0.3248	6.418	6.386	1.740	0.2036	0.1926	-1.005	1.775	0.6794
350S137-54	0.0566	50	0.3724	1.2691	0.6963	0.3979	1.367	0.0901	0.492	0.6845	0.3865	11.573	11.425	3.372	0.3977	0.2330	-0.991	1.759	0.6826
350S137-68	0.0713	50	0.4613	1.5722	0.8493	0.4853	1.357	0.1074	0.483	0.8506	0.4867	14.573	14.530	4.203	0.7817	0.2799	-0.973	1.738	0.6867
350S137-97	0.1017	50	0.6350	2.1642	1.1304	0.6460	1.334	0.1361	0.463	1.1347	0.6503	19.471	19.340	5.705	2.1893	0.3614	-0.935	1.694	0.6953
350S162-33	0.0346	33	0.2491	0.8491	0.4935	0.2820	1.407	0.0880	0.594	0.4572	0.2486	4.913	4.744	1.024	0.0994	0.2262	-1.235	1.964	0.6050
350S162-43	0.0451	33	0.3229	1.1003	0.6351	0.3629	1.403	0.1123	0.590	0.6180	0.3463	6.843	6.721	1.740	0.2189	0.2861	-1.223	1.952	0.6076
350S162-54	0.0566	50	0.4007	1.3655	0.7802	0.4458	1.395	0.1361	0.583	0.7517	0.4185	12.531	11.973	3.372	0.4279	0.3473	-1.208	1.936	0.6103
350S162-68	0.0713	50	0.4970	1.6937	0.9541	0.5452	1.386	0.1635	0.574	0.9337	0.5253	15.729	15.948	4.203	0.8421	0.4191	-1.190	1.914	0.6138
350S162-97	0.1017	50	0.6859	2.3375	1.2772	0.7298	1.365	0.2102	0.554	1.2816	0.7343	21.986	21.852	5.705	2.3646	0.5463	-1.151	1.869	0.6209
350S200-33	0.0346	33	0.2751	0.9375	0.5714	0.3265	1.441	0.1462	0.729	0.4973	0.2603	5.143	4.962	1.024	0.1098	0.3709	-1.570	2.252	0.5143
350S200-43	0.0451	33	0.3567	1.2156	0.7361	0.4206	1.437	0.1870	0.724	0.6670	0.3572	7.058	7.095	1.740	0.2418	0.4706	-1.557	2.239	0.5162
350S200-54	0.0566	50	0.4431	1.5102	0.9060	0.5177	1.430	0.2279	0.717	0.8098	0.4304	12.886	12.576	3.372	0.4732	0.5733	-1.543	2.222	0.5182
350S200-68	0.0713	50	0.5504	1.8759	1.1112	0.6350	1.421	0.2757	0.708	1.0529	0.5797	17.356	16.955	4.203	0.9328	0.6952	-1.523	2.200	0.5206
350S200-97	0.1017	50	0.7621	2.5974	1.4974	0.8557	1.402	0.3601	0.687	1.4793	0.8375	25.076	25.619	5.705	2.6276	0.9156	-1.483	2.153	0.5256
362S137-33	0.0346	33	0.2362	0.8049	0.4792	0.2644	1.424	0.0594	0.501	0.4696	0.2556	5.050	4.731	1.024	0.0942	0.1647	-1.003	1.813	0.6939

MEMBER	Design Thickness (in)	F _y (ksi)	GROSS PROPERTIES							EFFECTIVE PROPERTIES					TORSIONAL PROPERTIES				
			Area (in ²)	Weight (lb/ft)	I _x (in ⁴)	S _x (in ³)	r _x (in.)	I _y (in ⁴)	r _y (in.)	I _{xe} (in ⁴)	S _{xe} (in ³)	M _{al} (in-k)	M _{ad} (in-k)	V _{ay} (kip)	Jx1000 (in ⁴)	C _w (in ⁶)	X _o (in.)	r _o (in.)	β
362S137-43	0.0451	33	0.3059	1.0427	0.6163	0.3401	1.419	0.0755	0.497	0.6166	0.3404	6.726	6.655	1.740	0.2074	0.2077	-0.991	1.801	0.6970
362S137-54	0.0566	50	0.3795	1.2932	0.7560	0.4171	1.412	0.0911	0.490	0.7432	0.4053	12.134	11.911	3.372	0.4052	0.2514	-0.978	1.786	0.7002
362S137-68	0.0713	50	0.4702	1.6025	0.9227	0.5091	1.401	0.1087	0.481	0.9240	0.5105	15.284	15.242	4.370	0.7968	0.3021	-0.959	1.765	0.7044
362S137-97	0.1017	50	0.6477	2.2075	1.2296	0.6784	1.378	0.1376	0.461	1.2340	0.6828	20.442	20.312	5.943	2.2332	0.3904	-0.922	1.721	0.7130
362S162-33	0.0346	33	0.2535	0.8638	0.5349	0.2951	1.453	0.0891	0.593	0.4958	0.2605	5.148	4.932	1.024	0.1011	0.2438	-1.219	1.987	0.6234
362S162-43	0.0451	33	0.3285	1.1195	0.6886	0.3799	1.448	0.1136	0.588	0.6700	0.3626	7.164	6.997	1.740	0.2227	0.3085	-1.208	1.975	0.6261
362S162-54	0.0566	50	0.4078	1.3896	0.8461	0.4668	1.441	0.1377	0.581	0.8153	0.4384	13.125	12.468	3.372	0.4354	0.3746	-1.193	1.959	0.6288
362S162-68	0.0713	50	0.5059	1.7240	1.0353	0.5712	1.431	0.1654	0.572	1.0130	0.5504	16.479	16.632	4.370	0.8572	0.4523	-1.175	1.937	0.6324
362S162-97	0.1017	50	0.6986	2.3808	1.3874	0.7655	1.409	0.2127	0.552	1.3919	0.7699	23.052	22.919	5.943	2.4085	0.5902	-1.136	1.892	0.6397
362S200-33	0.0346	33	0.2794	0.9523	0.6185	0.3413	1.488	0.1479	0.728	0.5389	0.2726	5.388	5.152	1.024	0.1115	0.3997	-1.552	2.270	0.5323
362S200-43	0.0451	33	0.3623	1.2348	0.7970	0.4397	1.483	0.1893	0.723	0.7226	0.3739	7.389	7.374	1.740	0.2457	0.5073	-1.540	2.257	0.5343
362S200-54	0.0566	50	0.4502	1.5343	0.9813	0.5414	1.476	0.2307	0.716	0.8777	0.4508	13.498	13.077	3.372	0.4808	0.6183	-1.525	2.240	0.5364
362S200-68	0.0713	50	0.5594	1.9063	1.2041	0.6643	1.467	0.2790	0.706	1.1411	0.6068	18.168	17.652	4.370	0.9479	0.7502	-1.506	2.218	0.5389
362S200-97	0.1017	50	0.7749	2.6407	1.6242	0.8961	1.448	0.3646	0.686	1.6041	0.8768	26.253	26.829	5.943	2.6714	0.9891	-1.466	2.172	0.5442
400S137-33	0.0346	33	0.2491	0.8491	0.6031	0.3016	1.556	0.0612	0.496	0.5912	0.2917	5.764	5.287	0.976	0.0994	0.2036	-0.965	1.897	0.7412
400S137-43	0.0451	33	0.3229	1.1003	0.7764	0.3882	1.551	0.0778	0.491	0.7766	0.3885	7.677	7.472	1.740	0.2189	0.2571	-0.954	1.886	0.7442
400S137-54	0.0566	50	0.4007	1.3655	0.9534	0.4767	1.543	0.0939	0.484	0.9371	0.4633	13.871	13.380	3.372	0.4279	0.3114	-0.940	1.870	0.7473
400S137-68	0.0713	50	0.4970	1.6937	1.1653	0.5826	1.531	0.1121	0.475	1.1667	0.5840	17.485	17.455	4.871	0.8421	0.3748	-0.922	1.850	0.7514
400S137-97	0.1017	50	0.6859	2.3375	1.5580	0.7790	1.507	0.1420	0.455	1.5627	0.7832	23.450	23.324	6.658	2.3646	0.4858	-0.885	1.806	0.7598
400S162-33	0.0346	33	0.2664	0.9080	0.6711	0.3356	1.587	0.0919	0.587	0.6230	0.2973	5.874	5.500	0.976	0.1063	0.3015	-1.177	2.061	0.6741
400S162-43	0.0451	33	0.3454	1.1771	0.8645	0.4323	1.582	0.1173	0.583	0.8412	0.4128	8.157	7.830	1.740	0.2342	0.3818	-1.165	2.049	0.6768
400S162-54	0.0566	50	0.4290	1.4620	1.0634	0.5317	1.574	0.1422	0.576	1.0247	0.4998	14.964	13.964	3.372	0.4581	0.4641	-1.151	2.033	0.6796
400S162-68	0.0713	50	0.5326	1.8152	1.3029	0.6514	1.564	0.1708	0.566	1.2746	0.6278	18.798	18.712	4.871	0.9026	0.5611	-1.133	2.012	0.6833
400S162-97	0.1017	50	0.7367	2.5108	1.7512	0.8756	1.542	0.2198	0.546	1.7559	0.8799	26.345	26.216	6.658	2.5400	0.7343	-1.094	1.968	0.6908
400S200-33	0.0346	33	0.2924	0.9965	0.7732	0.3866	1.626	0.1529	0.723	0.6757	0.3110	6.145	5.727	0.976	0.1167	0.4941	-1.503	2.330	0.5836
400S200-43	0.0451	33	0.3792	1.2924	0.9968	0.4984	1.621	0.1957	0.718	0.9054	0.4257	8.411	8.221	1.740	0.2571	0.6278	-1.491	2.317	0.5857
400S200-54	0.0566	50	0.4714	1.6067	1.2284	0.6142	1.614	0.2385	0.711	1.1012	0.5140	15.390	14.592	3.372	0.5034	0.7659	-1.477	2.301	0.5879
400S200-68	0.0713	50	0.5861	1.9974	1.5092	0.7546	1.605	0.2886	0.702	1.4308	0.6905	20.672	19.769	4.871	0.9932	0.9305	-1.458	2.279	0.5908
400S200-97	0.1017	50	0.8130	2.7707	2.0410	1.0205	1.584	0.3773	0.681	2.0147	0.9980	29.881	30.554	6.658	2.8029	1.2304	-1.418	2.233	0.5967
600S137-33	0.0346	33	0.3183	1.0849	1.5826	0.5275	2.230	0.0685	0.464	1.4645	0.4541	8.974	8.187	0.638	0.1270	0.4999	-0.807	2.416	0.8885
600S137-43	0.0451	33	0.4131	1.4077	2.0424	0.6808	2.224	0.0871	0.459	1.9908	0.6450	12.745	11.824	1.415	0.2801	0.6329	-0.796	2.406	0.8904
600S137-54	0.0566	50	0.5139	1.7513	2.5183	0.8394	2.214	0.1051	0.452	2.4165	0.7767	23.255	21.248	2.822	0.5488	0.7693	-0.784	2.392	0.8925
600S137-68	0.0713	50	0.6396	2.1796	3.0954	1.0318	2.200	0.1255	0.443	3.0951	1.0317	30.889	28.903	5.352	1.0838	0.9299	-0.768	2.372	0.8952

MEMBER	Design Thickness (in)	F _y (ksi)	GROSS PROPERTIES							EFFECTIVE PROPERTIES					TORSIONAL PROPERTIES				
			Area (in ²)	Weight (lb/ft)	I _x (in ⁴)	S _x (in ³)	r _x (in.)	I _y (in ⁴)	r _y (in.)	I _{xe} (in ⁴)	S _{xe} (in ³)	M _{al} (in-k)	M _{ad} (in-k)	V _{ay} (kip)	Jx1000 (in ⁴)	C _w (in ⁶)	X _o (in.)	r _o (in.)	β
600S137-97	0.1017	50	0.8893	3.0306	4.1898	1.3966	2.171	0.1591	0.423	4.1954	1.4002	41.922	41.814	10.472	3.0659	1.2159	-0.734	2.330	0.9007
600S162-33	0.0346	33	0.3356	1.1439	1.7365	0.5788	2.275	0.1036	0.555	1.6058	0.5106	10.090	8.511	0.638	0.1339	0.7407	-0.996	2.544	0.8468
600S162-43	0.0451	33	0.4356	1.4845	2.2423	0.7474	2.269	0.1321	0.551	2.1846	0.7169	14.166	12.328	1.415	0.2953	0.9408	-0.985	2.534	0.8488
600S162-54	0.0566	50	0.5422	1.8478	2.7682	0.9227	2.260	0.1602	0.544	2.6730	0.8728	26.132	22.082	2.822	0.5790	1.1475	-0.972	2.519	0.8511
600S162-68	0.0713	50	0.6752	2.3011	3.4086	1.1362	2.247	0.1925	0.534	3.3363	1.0980	32.875	30.197	5.352	1.1442	1.3934	-0.955	2.499	0.8539
600S162-97	0.1017	50	0.9401	3.2039	4.6321	1.5440	2.220	0.2479	0.514	4.6377	1.5477	46.339	46.228	10.472	3.2412	1.8400	-0.920	2.457	0.8598
600S200-33	0.0346	33	0.3616	1.2323	1.9673	0.6558	2.333	0.1733	0.692	1.6959	0.5163	10.203	8.825	0.638	0.1443	1.2139	-1.292	2.755	0.7800
600S200-43	0.0451	33	0.4694	1.5998	2.5422	0.8474	2.327	0.2219	0.687	2.3305	0.7386	14.594	12.838	1.415	0.3183	1.5470	-1.281	2.744	0.7812
600S200-54	0.0566	50	0.5846	1.9924	3.1431	1.0477	2.319	0.2705	0.680	2.8488	0.8977	26.877	22.901	2.822	0.6243	1.8939	-1.267	2.729	0.7843
600S200-68	0.0713	50	0.7287	2.4834	3.8785	1.2928	2.307	0.3276	0.670	3.6907	1.1943	35.759	31.528	5.352	1.2348	2.3108	-1.249	2.708	0.7872
600S200-97	0.1017	50	1.0164	3.4639	5.2955	1.7652	2.283	0.4289	0.650	5.2224	1.7260	51.678	50.565	10.472	3.5042	3.0832	-1.212	2.665	0.7931
600S250-33 ⁶	0.0346	33	0.3962	1.3502	2.2752	0.7584	2.396	0.3024	0.874	--	--	--	--	--	0.1581	2.0795	-1.705	3.068	0.6912
600S250-43	0.0451	33	0.5145	1.7535	2.9420	0.9807	2.391	0.3884	0.869	2.5284	0.7722	15.260	13.243	1.415	0.3489	2.6582	-1.693	3.056	0.6931
600S250-54	0.0566	50	0.6412	2.1853	3.6429	1.2143	2.384	0.4758	0.861	3.0865	0.9364	28.037	23.527	2.822	0.6847	3.2652	-1.679	3.040	0.6951
600S250-68	0.0713	50	0.8000	2.7264	4.5051	1.5017	2.373	0.5800	0.851	3.9783	1.2334	36.927	32.611	5.352	1.3556	4.0014	-1.660	3.019	0.6976
600S250-97	0.1017	50	1.1181	3.8105	6.1800	2.0600	2.351	0.7708	0.830	5.8594	1.8909	56.612	53.258	10.472	3.8548	5.3882	-1.621	2.974	0.7029

¹Properties are based on the full, unreduced cross-section away from any web punch-outs.

²Use the effective moment of inertia for deflection calculation.

³Allowable moment is lesser of M_{al} and M_{ad}. Distortional buckling is based on an assumed KΦ = 0.

⁴The effect of cold-work of forming on strength was conservatively not incorporated.

⁵See technical manual for product identification information. Calculations were performed using dimensions specified in the client's technical manual dated 7/29/2025.

⁶Flange width-to-thickness ratio exceeds 60. Allowable design values for these members are outside the scope and may be determined in accordance with AISI S100.

SYMBOLS

I_x = Moment of inertia about x-axis
S_x = Section modulus about x-axis
r_x = Radius of gyration about x-axis
I_y = Moment of inertia about y-axis
r_y = Radius of gyration about y-axis

M_{al} = Allowable bending moment about x-axis based on local buckling
M_{ad} = Allowable moment about x-axis based on distortional buckling
V_{ay} = Allowable shear

J = St. Venant torsion constant
C_w = Torsional warping constant
x_o = Distance from shear center to the centroid along the principal X-axis
r_o = Polar radius of gyration about shear center
β = Torsional flexural constant

Section Properties Document Pages 8-9 below contain the section properties for the Joists.

NR-25-03-17 JOIST SECTION PROPERTIES^{1,2}

MEMBER	Design Thickness (in)	Hole Height (in.)	F _y (ksi)	GROSS PROPERTIES							NET PROPERTIES (AT HOLE) ⁴						GROSS TORSIONAL PROPERTIES				
				Area (in ²)	Weight (lb/ft)	I _x (in ⁴)	S _x (in ³)	r _x (in.)	I _y (in ⁴)	r _y (in.)	Area (in ²)	I _{x,net} (in ⁴)	S _{x,net} (in ³)	r _{x,net} (in.)	I _{y,net} (in ⁴)	r _{y,net} (in.)	Jx1000 (in ⁴)	C _w (in ⁶)	X _o (in.)	r _o (in.)	β ³
725J175-43	0.0451	4.25	33	0.5258	1.7919	3.9013	1.0762	2.724	0.2062	0.626	0.33413	3.6120	0.9964	3.2879	0.1547	0.6803	0.3565	2.1757	-1.1339	3.0160	0.8587
725J175-54	0.0566	4.25	50	0.6554	2.2336	4.8298	1.3324	2.715	0.2517	0.620	0.41484	4.4669	1.2323	3.2815	0.1886	0.6743	0.6999	2.6430	-1.1202	3.0013	0.8607
725J175-68	0.0713	4.25	50	0.8178	2.7871	5.9707	1.6471	2.702	0.3054	0.611	0.51479	5.5140	1.5211	3.2728	0.2286	0.6664	1.3858	3.1845	-1.1021	2.9814	0.8634
725J175-97	0.1017	4.25	50	1.1435	3.8971	8.1836	2.2576	2.675	0.4019	0.593	0.71130	7.5337	2.0783	3.2544	0.3001	0.6495	3.9424	4.1261	-1.0641	2.9397	0.8690
800J175-43	0.0451	4.25	33	0.5596	1.9072	4.9465	1.2366	2.973	0.2116	0.615	0.36796	4.6568	1.1642	3.5575	0.1677	0.6751	0.3794	2.7007	-1.0817	3.2226	0.8873
800J175-54	0.0566	4.25	50	0.6978	2.3782	6.1279	1.5320	2.963	0.2584	0.608	0.45729	5.7646	1.4412	3.5505	0.2047	0.6690	0.7452	3.2828	-1.0683	3.2080	0.8891
800J175-68	0.0713	4.25	50	0.8713	2.9694	7.5825	1.8956	2.950	0.3134	0.600	0.56827	7.1253	1.7813	3.5410	0.2482	0.6609	1.4765	3.9588	-1.0505	3.1883	0.8914
800J175-97	0.1017	4.25	50	1.2198	4.1571	10.4137	2.6034	2.922	0.4124	0.581	0.78757	9.763	2.4408	3.5209	0.326	0.6438	4.2054	5.1392	-1.0133	3.1469	0.8963
925J175-43	0.0451	4.25	33	0.6160	2.0993	7.0431	1.5228	3.381	0.2194	0.597	0.42433	6.7527	1.4600	3.9892	0.1848	0.6599	0.4177	3.7262	-1.0055	3.5774	0.9210
925J175-54	0.0566	4.25	50	0.7686	2.6193	8.7337	1.8884	3.371	0.2678	0.590	0.52804	8.3695	1.8096	3.9812	0.2256	0.6537	0.8207	4.5329	-0.9925	3.5629	0.9224
925J175-68	0.0713	4.25	50	0.9604	3.2731	10.8212	2.3397	3.357	0.3249	0.582	0.65739	10.363	2.2406	3.9703	0.274	0.6454	1.6275	5.4724	-0.9753	3.5433	0.9242
925J175-97	0.1017	4.25	50	1.3469	4.5903	14.9039	3.2225	3.326	0.4274	0.563	0.91470	14.252	3.0815	3.9473	0.361	0.6279	4.6437	7.1217	-0.9393	3.5021	0.9281
925J175-43	0.0451	6.25	33	0.6160	2.0993	7.0431	1.5228	3.381	0.2194	0.597	0.33413	6.1237	1.3240	4.2810	0.1547	0.6803	0.4177	3.7262	-1.0055	3.5774	0.9210
925J175-54	0.0566	6.25	50	0.7686	2.6193	8.7337	1.8884	3.371	0.2678	0.590	0.41484	7.5801	1.6389	4.2746	0.1886	0.6743	0.8207	4.5329	-0.9925	3.5629	0.9224
925J175-68	0.0713	6.25	50	0.9604	3.2731	10.8212	2.3397	3.357	0.3249	0.582	0.51479	9.368	2.0256	4.2660	0.229	0.6664	1.6275	5.4724	-0.9753	3.5433	0.9242
925J175-97	0.1017	6.25	50	1.3469	4.5903	14.9039	3.2225	3.326	0.4274	0.563	0.71130	12.834	2.7748	4.2477	0.300	0.6495	4.6437	7.1217	-0.9393	3.5021	0.9281
1000J200-54	0.0566	4.25	50	0.8393	2.8605	11.2816	2.2563	3.666	0.3777	0.671	0.59879	10.917	2.1833	4.2698	0.324	0.7361	0.8963	7.3734	-1.1289	3.8939	0.9159
1000J200-68	0.0713	4.25	50	1.0495	3.5768	13.9988	2.7998	3.652	0.4599	0.662	0.74652	13.540	2.7079	4.2588	0.395	0.7275	1.7785	8.9378	-1.1114	3.8742	0.9177
1000J200-97	0.1017	4.25	50	1.4740	5.0236	19.3425	3.8685	3.622	0.6099	0.643	1.0418	18.690	3.7380	4.2355	0.525	0.7096	5.0820	11.737	-1.0747	3.8328	0.9214
1000J200-54	0.0566	6.25	50	0.8393	2.8605	11.2816	2.2563	3.666	0.3777	0.671	0.48559	10.127	2.0254	4.5668	0.281	0.7605	0.8963	7.3734	-1.1289	3.8939	0.9159
1000J200-68	0.0713	6.25	50	1.0495	3.5768	13.9988	2.7998	3.652	0.4599	0.662	0.60392	12.545	2.5090	4.5578	0.342	0.7523	1.7785	8.9378	-1.1114	3.8742	0.9177
1000J200-97	0.1017	6.25	50	1.4740	5.0236	19.3425	3.8685	3.622	0.6099	0.643	0.83842	17.271	3.4543	4.5387	0.453	0.7349	5.0820	11.737	-1.0747	3.8328	0.9214
1125J175-54	0.0566	4.25	50	0.8818	3.0051	14.1756	2.5201	4.009	0.2797	0.563	0.64124	13.809	2.4550	4.6406	0.250	0.6238	0.9416	7.0284	-0.8926	4.1455	0.9536
1125J175-68	0.0713	4.25	50	1.1030	3.7591	17.5935	3.1277	3.994	0.3393	0.555	0.79999	17.133	3.0458	4.6277	0.303	0.6153	1.8691	8.4954	-0.8763	4.1258	0.9549
1125J175-97	0.1017	4.25	50	1.5503	5.2835	24.3195	4.3235	3.961	0.4462	0.536	1.1181	23.665	4.2070	4.6005	0.399	0.5976	5.3449	11.086	-0.8423	4.0844	0.9575
1125J175-54	0.0566	6.25	50	0.8818	3.0051	14.1756	2.5201	4.009	0.2797	0.563	0.52804	13.020	2.3146	4.9656	0.226	0.6537	0.9416	7.0284	-0.8926	4.1455	0.9536
1125J175-68	0.0713	6.25	50	1.1030	3.7591	17.5935	3.1277	3.994	0.3393	0.555	0.65739	16.138	2.8690	4.9547	0.274	0.6454	1.8691	8.4954	-0.8763	4.1258	0.9549
1125J175-97	0.1017	6.25	50	1.5503	5.2835	24.3195	4.3235	3.961	0.4462	0.536	0.91470	22.246	3.9548	4.9316	0.361	0.6279	5.3449	11.086	-0.8423	4.0844	0.9575
1200J200-54	0.0566	4.25	50	0.9525	3.2462	17.6683	2.9447	4.307	0.3935	0.643	0.71199	17.301	2.8835	4.9295	0.354	0.7053	1.0172	11.119	-1.0246	4.4728	0.9475
1200J200-68	0.0713	4.25	50	1.1921	4.0628	21.9550	3.6592	4.291	0.4791	0.634	0.88912	21.493	3.5822	4.9166	0.431	0.6966	2.0202	13.492	-1.0078	4.4530	0.9488
1200J200-97	0.1017	4.25	50	1.6774	5.7167	30.4280	5.0713	4.259	0.6352	0.615	1.2452	29.771	4.9619	4.8896	0.573	0.6783	5.7832	17.759	-0.9729	4.4116	0.9514

MEMBER	Design Thickness (in)	Hole Height (in.)	F _y (ksi)	GROSS PROPERTIES							NET PROPERTIES (AT HOLE) ⁴						GROSS TORSIONAL PROPERTIES				
				Area (in ²)	Weight (lb/ft)	I _x (in ⁴)	S _x (in ³)	r _x (in.)	I _y (in ⁴)	r _y (in.)	Area (in ²)	I _{x,net} (in ⁴)	S _{x,net} (in ³)	r _{x,net} (in.)	I _{y,net} (in ⁴)	r _{y,net} (in.)	Jx1000 (in ⁴)	C _w (in ⁶)	x _o (in.)	r _o (in.)	β ³
1200J200-54	0.0566	8.00	50	0.9525	3.2462	17.6683	2.9447	4.307	0.3935	0.643	0.49974	15.248	2.5413	5.5238	0.287	0.7584	1.0172	11.119	-1.0246	4.4728	0.9475
1200J200-68	0.0713	8.00	50	1.1921	4.0628	21.9550	3.6592	4.291	0.4791	0.634	0.62174	18.907	3.1512	5.5145	0.350	0.7501	2.0202	13.492	-1.0078	4.4530	0.9488
1200J200-97	0.1017	8.00	50	1.6774	5.7167	30.4280	5.0713	4.259	0.6352	0.615	0.86385	26.083	4.3471	5.4949	0.464	0.7326	5.7832	17.759	-0.9729	4.4116	0.9514
1400J200-68	0.0713	4.25	50	1.3347	4.5488	32.2968	4.6138	4.919	0.4942	0.608	1.0317	31.830	4.5472	5.554	0.458	0.666	2.2618	19.128	-0.923	5.041	0.9665
1400J200-97	0.1017	4.25	50	1.8808	6.4099	44.8701	6.4100	4.884	0.6550	0.590	1.4486	44.208	6.3154	5.524	0.608	0.648	6.4845	25.215	-0.890	4.999	0.9683
1400J200-68	0.0713	10.00	50	1.3347	4.5488	32.2968	4.6138	4.919	0.4942	0.608	0.62174	26.345	3.7635	6.509	0.350	0.750	2.2618	19.128	-0.923	5.041	0.9665
1400J200-97	0.1017	10.00	50	1.8808	6.4099	44.8701	6.4100	4.884	0.6550	0.590	0.86385	36.383	5.1976	6.490	0.464	0.733	6.4845	25.215	-0.890	4.999	0.9683

¹Gross properties are based on the full, unreduced cross-section away from any web punch-outs. Net properties are based on the cross-section at the hole or punch-out.

²See technical manual for product identification information. Calculations were performed using dimensions specified in the client's technical manual dated 7/29/2025.

³ $\beta = 1 - (x_o/r_o)^2$

⁴Net properties are conservatively ignoring any lips at the hole or punch-out.

SYMBOLS (if net is added, it means property at hole)

I_x = Moment of inertia about x-axis

S_x = Section modulus about x-axis

r_x = Radius of gyration about x-axis

I_y = Moment of inertia about y-axis

r_y = Radius of gyration about y-axis

J = St. Venant torsion constant

C_w = Torsional warping constant

x_o = Distance from shear center to the centroid along the principal X-axis

r_o = Polar radius of gyration about shear center

β = Torsional flexural constant

Section Properties Document Pages 11-15 below contain the section properties for the Tracks.

NR-25-03-17 TRACK SECTION PROPERTIES^{1,2,3}

MEMBER	Design Thickness (in)	F _y (ksi)	GROSS PROPERTIES							EFFECTIVE PROPERTIES				TORSIONAL PROPERTIES				
			Area (in ²)	Weight (lb/ft)	I _x (in ⁴)	S _x (in ³)	r _x (in.)	I _y (in ⁴)	r _y (in.)	I _{xe} (in ⁴)	S _{xe} (in ³)	M _{al} (in-k)	V _{ay} (kip)	Jx1000 (in ⁴)	C _w (in ⁶)	X _o (in.)	r _o (in.)	β
162T125-18 ⁴	0.0188	50	0.0753	0.2567	0.0356	0.0438	0.687	0.0128	0.412	--	--	--	--	0.0089	0.0059	-0.895	1.201	0.4451
162T125-27	0.0283	33	0.1129	0.3846	0.0527	0.0648	0.683	0.0190	0.411	0.0383	0.0393	0.777	0.494	0.0301	0.0087	-0.892	1.196	0.4442
162T125-30	0.0312	33	0.1242	0.4234	0.0578	0.0711	0.682	0.0209	0.410	0.0432	0.0450	0.889	0.543	0.0403	0.0095	-0.891	1.195	0.4439
162T125-33	0.0346	33	0.1375	0.4688	0.0637	0.0784	0.681	0.0231	0.410	0.0491	0.0519	1.026	0.601	0.0549	0.0105	-0.890	1.193	0.4435
250T125-33	0.0346	33	0.1678	0.5719	0.1685	0.1348	1.002	0.0267	0.399	0.1344	0.0950	1.876	0.975	0.0670	0.0287	-0.776	1.329	0.6590
250T125-43	0.0451	33	0.2178	0.7423	0.2168	0.1735	0.998	0.0344	0.397	0.1849	0.1351	2.669	1.265	0.1477	0.0367	-0.773	1.323	0.6590
250T125-54	0.0566	50	0.2711	0.9239	0.2668	0.2134	0.992	0.0424	0.396	0.2306	0.1696	5.079	2.353	0.2895	0.0449	-0.770	1.317	0.6579
250T125-68	0.0713	50	0.3376	1.1506	0.3272	0.2617	0.984	0.0523	0.393	0.3024	0.2308	6.909	2.866	0.5721	0.0549	-0.767	1.309	0.6562
250T125-97	0.1017	50	0.4701	1.6020	0.4408	0.3526	0.968	0.0711	0.389	0.4435	0.3558	10.653	3.798	1.6206	0.0735	-0.762	1.292	0.6522
250T150-33	0.0346	33	0.1851	0.6309	0.1948	0.1558	1.026	0.0439	0.487	0.1447	0.0985	1.947	0.975	0.0739	0.0472	-0.992	1.508	0.5674
250T150-43	0.0451	33	0.2404	0.8191	0.2508	0.2007	1.022	0.0566	0.485	0.2002	0.1409	2.784	1.265	0.1630	0.0603	-0.988	1.502	0.5670
250T150-54	0.0566	50	0.2994	1.0203	0.3090	0.2472	1.016	0.0700	0.484	0.2501	0.1772	5.306	2.353	0.3197	0.0741	-0.986	1.496	0.5656
250T150-68	0.0713	50	0.3733	1.2721	0.3797	0.3038	1.009	0.0865	0.481	0.3306	0.2433	7.285	2.866	0.6325	0.0908	-0.984	1.489	0.5634
250T150-97	0.1017	50	0.5209	1.7753	0.5139	0.4111	0.993	0.1183	0.477	0.4981	0.3902	11.682	3.798	1.7959	0.1223	-0.979	1.474	0.5587
250T200-33	0.0346	33	0.2197	0.7488	0.2474	0.1979	1.061	0.0955	0.659	0.1622	0.1037	2.049	0.975	0.0877	0.1035	-1.440	1.907	0.4293
250T200-43	0.0451	33	0.2855	0.9728	0.3188	0.2550	1.057	0.1235	0.658	0.2260	0.1492	2.948	1.265	0.1935	0.1326	-1.437	1.901	0.4286
250T200-54	0.0566	50	0.3560	1.2132	0.3935	0.3148	1.051	0.1531	0.656	0.2828	0.1880	5.628	2.353	0.3801	0.1633	-1.435	1.896	0.4270
250T200-68	0.0713	50	0.4446	1.5151	0.4849	0.3879	1.044	0.1898	0.653	0.3778	0.2607	7.806	2.866	0.7533	0.2009	-1.434	1.890	0.4247
250T200-97	0.1017	50	0.6226	2.1219	0.6601	0.5281	1.030	0.2616	0.648	0.5829	0.4302	12.881	3.798	2.1465	0.2726	-1.430	1.878	0.4199
350T125-33	0.0346	33	0.2024	0.6899	0.3677	0.2101	1.348	0.0295	0.382	0.3016	0.1556	3.075	1.024	0.0808	0.0635	-0.680	1.557	0.8095
350T125-43	0.0451	33	0.2629	0.8960	0.4745	0.2711	1.343	0.0380	0.380	0.4121	0.2184	4.315	1.740	0.1782	0.0812	-0.676	1.551	0.8099
350T125-54	0.0566	50	0.3277	1.1168	0.5860	0.3349	1.337	0.0470	0.379	0.5148	0.2744	8.217	3.372	0.3499	0.0999	-0.674	1.545	0.8097
350T125-68	0.0713	50	0.4089	1.3936	0.7222	0.4127	1.329	0.0580	0.377	0.6726	0.3694	11.061	4.203	0.6929	0.1226	-0.671	1.535	0.8093
350T125-97	0.1017	50	0.5718	1.9486	0.9837	0.5621	1.312	0.0792	0.372	0.9868	0.5648	16.911	5.705	1.9712	0.1656	-0.664	1.517	0.8083
350T150-33	0.0346	33	0.2197	0.7488	0.4196	0.2398	1.382	0.0488	0.471	0.3233	0.1614	3.189	1.024	0.0877	0.1039	-0.880	1.705	0.7336
350T150-43	0.0451	33	0.2855	0.9728	0.5417	0.3096	1.378	0.0630	0.470	0.4435	0.2276	4.497	1.740	0.1935	0.1332	-0.877	1.699	0.7338
350T150-54	0.0566	50	0.3560	1.2132	0.6699	0.3828	1.372	0.0780	0.468	0.5548	0.2864	8.574	3.372	0.3801	0.1642	-0.874	1.693	0.7333
350T150-68	0.0713	50	0.4446	1.5151	0.8270	0.4726	1.364	0.0965	0.466	0.7297	0.3887	11.636	4.203	0.7533	0.2020	-0.871	1.684	0.7325
350T150-97	0.1017	50	0.6226	2.1219	1.1306	0.6460	1.348	0.1327	0.462	1.0964	0.6151	18.415	5.705	2.1465	0.2745	-0.865	1.666	0.7306
350T200-33	0.0346	33	0.2543	0.8667	0.5235	0.2991	1.435	0.1070	0.649	0.3607	0.1702	3.363	1.024	0.1015	0.2263	-1.303	2.044	0.5936
350T200-43	0.0451	33	0.3306	1.1265	0.6763	0.3865	1.430	0.1383	0.647	0.4976	0.2412	4.766	1.740	0.2241	0.2908	-1.299	2.038	0.5934

MEMBER	Design Thickness (in)	F _y (ksi)	GROSS PROPERTIES							EFFECTIVE PROPERTIES				TORSIONAL PROPERTIES				
			Area (in ²)	Weight (lb/ft)	I _x (in ⁴)	S _x (in ³)	r _x (in.)	I _y (in ⁴)	r _y (in.)	I _{xe} (in ⁴)	S _{xe} (in ³)	M _{al} (in-k)	V _{ay} (kip)	Jx1000 (in ⁴)	C _w (in ⁶)	X _o (in.)	r _o (in.)	β
350T200-54	0.0566	50	0.4126	1.4061	0.8377	0.4787	1.425	0.1718	0.645	0.6235	0.3039	9.100	3.372	0.4406	0.3593	-1.297	2.032	0.5925
350T200-68	0.0713	50	0.5159	1.7580	1.0366	0.5923	1.418	0.2134	0.643	0.8270	0.4162	12.460	4.203	0.8742	0.4437	-1.295	2.025	0.5911
350T200-97	0.1017	50	0.7243	2.4685	1.4242	0.8183	1.402	0.2954	0.639	1.2683	0.6757	20.229	5.705	2.4972	0.6072	-1.289	2.009	0.5881
362T125-33	0.0346	33	0.2067	0.7046	0.3993	0.2203	1.390	0.0298	0.380	0.3285	0.1640	3.241	1.024	0.0825	0.0689	-0.669	1.589	0.8224
362T125-43	0.0451	33	0.2685	0.9152	0.5154	0.2844	1.385	0.0384	0.378	0.4486	0.2299	4.543	1.740	0.1821	0.0882	-0.666	1.583	0.8229
362T125-54	0.0566	50	0.3348	1.1409	0.6368	0.3514	1.379	0.0474	0.376	0.5605	0.2889	8.650	3.372	0.3575	0.1085	-0.663	1.576	0.8228
362T125-68	0.0713	50	0.4178	1.4239	0.7853	0.4333	1.371	0.0586	0.374	0.7320	0.3885	11.631	4.370	0.7080	0.1333	-0.660	1.567	0.8225
362T125-97	0.1017	50	0.5845	1.9919	1.0707	0.5907	1.353	0.0801	0.370	1.0738	0.5934	17.765	5.943	2.0151	0.1802	-0.654	1.548	0.8217
362T150-33	0.0346	33	0.2240	0.7636	0.4551	0.2511	1.425	0.0493	0.469	0.3519	0.1701	3.362	1.024	0.0894	0.1128	-0.868	1.733	0.7493
362T150-43	0.0451	33	0.2911	0.9920	0.5877	0.3242	1.421	0.0636	0.468	0.4825	0.2395	4.733	1.740	0.1974	0.1447	-0.865	1.728	0.7496
362T150-54	0.0566	50	0.3631	1.2373	0.7269	0.4011	1.415	0.0788	0.466	0.6036	0.3014	9.024	3.372	0.3877	0.1784	-0.862	1.721	0.7492
362T150-68	0.0713	50	0.4535	1.5454	0.8978	0.4954	1.407	0.0976	0.464	0.7934	0.4086	12.232	4.370	0.7684	0.2196	-0.859	1.713	0.7485
362T150-97	0.1017	50	0.6353	2.1652	1.2285	0.6778	1.391	0.1342	0.460	1.1916	0.6456	19.330	5.943	2.1904	0.2986	-0.853	1.695	0.7468
362T200-33	0.0346	33	0.2586	0.8815	0.5666	0.3126	1.480	0.1082	0.647	0.3924	0.1794	3.545	1.024	0.1032	0.2456	-1.288	2.066	0.6115
362T200-43	0.0451	33	0.3362	1.1457	0.7322	0.4040	1.476	0.1399	0.645	0.5408	0.2539	5.017	1.740	0.2279	0.3157	-1.284	2.060	0.6113
362T200-54	0.0566	50	0.4197	1.4302	0.9071	0.5005	1.470	0.1738	0.643	0.6778	0.3199	9.578	3.372	0.4481	0.3902	-1.282	2.054	0.6105
362T200-68	0.0713	50	0.5248	1.7884	1.1229	0.6169	1.463	0.2158	0.641	0.8982	0.4374	13.096	4.370	0.8893	0.4820	-1.279	2.046	0.6092
362T200-97	0.1017	50	0.7370	2.5118	1.5441	0.8519	1.447	0.2989	0.637	1.3765	0.7088	21.223	5.943	2.5410	0.6601	-1.274	2.031	0.6064
400T125-33	0.0346	33	0.2197	0.7488	0.5041	0.2521	1.515	0.0306	0.373	0.4183	0.1904	3.763	0.976	0.0877	0.0869	-0.641	1.686	0.8557
400T125-43	0.0451	33	0.2855	0.9728	0.6512	0.3256	1.510	0.0394	0.372	0.5699	0.2659	5.253	1.740	0.1935	0.1113	-0.638	1.681	0.8562
400T125-54	0.0566	50	0.3560	1.2132	0.8053	0.4026	1.504	0.0487	0.370	0.7142	0.3341	10.004	3.372	0.3801	0.1370	-0.635	1.674	0.8562
400T125-68	0.0713	50	0.4446	1.5151	0.9943	0.4971	1.495	0.0602	0.368	0.9290	0.4479	13.409	4.871	0.7533	0.1684	-0.631	1.665	0.8561
400T125-97	0.1017	50	0.6226	2.1219	1.3593	0.6797	1.478	0.0823	0.364	1.3625	0.6821	20.424	6.658	2.1465	0.2281	-0.625	1.645	0.8558
400T150-33	0.0346	33	0.2370	0.8078	0.5721	0.2861	1.554	0.0508	0.463	0.4473	0.1974	3.901	0.976	0.0946	0.1422	-0.834	1.823	0.7908
400T150-43	0.0451	33	0.3080	1.0497	0.7393	0.3697	1.549	0.0655	0.461	0.6116	0.2768	5.470	1.740	0.2088	0.1825	-0.831	1.817	0.7911
400T150-54	0.0566	50	0.3843	1.3097	0.9153	0.4577	1.543	0.0811	0.459	0.7655	0.3483	10.430	3.372	0.4104	0.2252	-0.828	1.811	0.7909
400T150-68	0.0713	50	0.4802	1.6365	1.1318	0.5659	1.535	0.1005	0.457	1.0045	0.4705	14.088	4.871	0.8137	0.2774	-0.825	1.802	0.7905
400T150-97	0.1017	50	0.6735	2.2952	1.5525	0.7762	1.518	0.1383	0.453	1.5068	0.7405	22.170	6.658	2.3218	0.3780	-0.818	1.783	0.7894
400T200-33	0.0346	33	0.2716	0.9257	0.7082	0.3541	1.615	0.1116	0.641	0.4978	0.2082	4.115	0.976	0.1084	0.3092	-1.244	2.137	0.6609
400T200-43	0.0451	33	0.3531	1.2034	0.9157	0.4579	1.610	0.1444	0.639	0.6841	0.2934	5.798	1.740	0.2394	0.3977	-1.241	2.131	0.6609
400T200-54	0.0566	50	0.4409	1.5026	1.1353	0.5677	1.605	0.1793	0.638	0.8574	0.3697	11.068	3.372	0.4708	0.4919	-1.239	2.125	0.6603
400T200-68	0.0713	50	0.5515	1.8795	1.4069	0.7035	1.597	0.2229	0.636	1.1339	0.5035	15.075	4.871	0.9346	0.6082	-1.236	2.117	0.6593
400T200-97	0.1017	50	0.7752	2.6418	1.9389	0.9694	1.582	0.3089	0.631	1.7338	0.8117	24.303	6.658	2.6725	0.8345	-1.230	2.101	0.6571

MEMBER	Design Thickness (in)	F _y (ksi)	GROSS PROPERTIES							EFFECTIVE PROPERTIES				TORSIONAL PROPERTIES				
			Area (in ²)	Weight (lb/ft)	I _x (in ⁴)	S _x (in ³)	r _x (in.)	I _y (in ⁴)	r _y (in.)	I _{xe} (in ⁴)	S _{xe} (in ³)	M _{al} (in-k)	V _{ay} (kip)	Jx1000 (in ⁴)	C _w (in ⁶)	X _o (in.)	r _o (in.)	β
550T125-33	0.0346	33	0.2716	0.9257	1.0871	0.3953	2.001	0.0330	0.349	0.8391	0.2617	5.172	0.698	0.1084	0.1835	-0.548	2.103	0.9321
550T125-43	0.0451	33	0.3531	1.2034	1.4068	0.5116	1.996	0.0425	0.347	1.2086	0.4023	7.950	1.550	0.2394	0.2354	-0.545	2.098	0.9325
550T125-54	0.0566	50	0.4409	1.5026	1.7445	0.6343	1.989	0.0526	0.345	1.5234	0.5123	15.337	3.093	0.4708	0.2903	-0.542	2.090	0.9372
550T125-68	0.0713	50	0.5515	1.8795	2.1617	0.7861	1.980	0.0650	0.343	2.0371	0.7192	21.532	5.352	0.9346	0.3576	-0.539	2.080	0.9329
550T125-97	0.1017	50	0.7752	2.6418	2.9786	1.0831	1.960	0.0891	0.339	2.9819	1.0852	32.490	9.518	2.6725	0.4868	-0.532	2.059	0.9333
550T150-33	0.0346	33	0.2889	0.9846	1.2162	0.4423	2.052	0.0551	0.437	0.9456	0.3027	5.982	0.698	0.1153	0.3006	-0.723	2.219	0.8939
550T150-43	0.0451	33	0.3757	1.2802	1.5745	0.5726	2.047	0.0711	0.435	1.3358	0.4479	8.850	1.550	0.2547	0.3864	-0.720	2.213	0.8943
550T150-54	0.0566	50	0.4692	1.5990	1.9541	0.7106	2.041	0.0882	0.434	1.6731	0.5634	16.869	3.093	0.5010	0.4776	-0.717	2.206	0.8944
550T150-68	0.0713	50	0.5872	2.0010	2.4244	0.8816	2.032	0.1093	0.432	2.1832	0.7526	22.534	5.352	0.9950	0.5899	-0.713	2.196	0.8945
550T150-97	0.1017	50	0.8260	2.8150	3.3490	1.2178	2.014	0.1508	0.427	3.2588	1.1680	34.970	9.518	2.8478	0.8077	-0.706	2.176	0.8947
550T200-33	0.0346	33	0.3235	1.1026	1.4746	0.5362	2.135	0.1226	0.616	1.0027	0.2992	5.912	0.698	0.1291	0.6526	-1.100	2.479	0.8032
550T200-43	0.0451	33	0.4208	1.4339	1.9100	0.6946	2.131	0.1585	0.614	1.4806	0.4740	9.366	1.550	0.2853	0.8409	-1.097	2.474	0.8035
550T200-54	0.0566	50	0.5258	1.7919	2.3734	0.8630	2.125	0.1971	0.612	1.8568	0.5969	17.872	3.093	0.5615	1.0420	-1.094	2.467	0.8034
550T200-68	0.0713	50	0.6585	2.2440	2.9497	1.0726	2.117	0.2452	0.610	2.4378	0.8030	24.041	5.352	1.1158	1.2915	-1.091	2.458	0.8031
550T200-97	0.1017	50	0.9277	3.1616	4.0900	1.4873	2.100	0.3407	0.606	3.6981	1.2725	38.098	9.518	3.1984	1.7808	-1.084	2.439	0.8026
600T125-33	0.0346	33	0.2889	0.9846	1.3464	0.4488	2.159	0.0336	0.341	1.0226	0.2891	5.713	0.638	0.1153	0.2248	-0.523	2.247	0.9458
600T125-43	0.0451	33	0.3757	1.2802	1.7432	0.5811	2.154	0.0433	0.339	1.4799	0.4470	8.832	1.415	0.2547	0.2885	-0.520	2.242	0.9461
600T125-54	0.0566	50	0.4692	1.5990	2.1631	0.7210	2.147	0.0536	0.338	1.8677	0.5699	17.063	2.822	0.5010	0.3560	-0.518	2.234	0.9463
600T125-68	0.0713	50	0.5872	2.0010	2.6829	0.8943	2.138	0.0662	0.336	2.5235	0.8147	24.391	5.352	0.9950	0.4387	-0.514	2.224	0.9465
600T125-97	0.1017	50	0.8260	2.8150	3.7037	1.2346	2.118	0.0908	0.332	3.7070	1.2365	37.021	10.472	2.8478	0.5978	-0.507	2.203	0.9470
600T150-33	0.0346	33	0.3062	1.0436	1.5003	0.5001	2.213	0.0563	0.429	1.1358	0.3276	6.474	0.638	0.1222	0.3686	-0.692	2.358	0.9138
600T150-43	0.0451	33	0.3982	1.3571	1.9431	0.6477	2.209	0.0726	0.427	1.6599	0.5126	10.129	1.415	0.2700	0.4740	-0.689	2.353	0.9142
600T150-54	0.0566	50	0.4975	1.6955	2.4130	0.8043	2.202	0.0900	0.425	2.0794	0.6448	19.304	2.822	0.5312	0.5860	-0.686	2.346	0.9144
600T150-68	0.0713	50	0.6228	2.1225	2.9961	0.9987	2.193	0.1116	0.423	2.7092	0.8587	25.711	5.352	1.0554	0.7242	-0.683	2.336	0.9145
600T150-97	0.1017	50	0.8769	2.9883	4.1460	1.3820	2.174	0.1541	0.419	4.0376	1.3276	39.748	10.472	3.0231	0.9926	-0.676	2.315	0.9148
600T200-33	0.0346	33	0.3408	1.1615	1.8081	0.6027	2.303	0.1255	0.607	1.2053	0.3258	6.437	0.638	0.1360	0.8005	-1.059	2.607	0.8348
600T200-43	0.0451	33	0.4433	1.5108	2.3430	0.7810	2.299	0.1623	0.605	1.8346	0.5420	10.711	1.415	0.3006	1.0318	-1.056	2.601	0.8351
600T200-54	0.0566	50	0.5541	1.8883	2.9128	0.9709	2.293	0.2018	0.603	2.3009	0.6825	20.435	2.822	0.5917	1.2791	-1.053	2.594	0.8351
600T200-68	0.0713	50	0.6941	2.3655	3.6227	1.2076	2.285	0.2511	0.601	3.0152	0.9151	27.400	5.352	1.1762	1.5862	-1.050	2.585	0.8350
600T200-97	0.1017	50	0.9786	3.3349	5.0305	1.6768	2.267	0.3491	0.597	4.5633	1.4434	43.216	10.472	3.3737	2.1897	-1.043	2.566	0.8347
800T125-33 ⁵	0.0346	33	0.3581	1.2205	2.7669	0.6917	2.780	0.0355	0.315	1.9585	0.3987	7.879	0.474	0.1429	0.4379	-0.444	2.832	0.9755
800T125-43	0.0451	33	0.4659	1.5876	3.5870	0.8967	2.775	0.0457	0.313	2.8795	0.6261	12.371	1.051	0.3158	0.5625	-0.441	2.827	0.9757
800T125-54	0.0566	50	0.5824	1.9848	4.4598	1.1149	2.767	0.0565	0.312	3.6477	0.8013	23.990	2.091	0.6219	0.6947	-0.438	2.819	0.9758

MEMBER	Design Thickness (in)	F _y (ksi)	GROSS PROPERTIES							EFFECTIVE PROPERTIES				TORSIONAL PROPERTIES				
			Area (in ²)	Weight (lb/ft)	I _x (in ⁴)	S _x (in ³)	r _x (in.)	I _y (in ⁴)	r _y (in.)	I _{xe} (in ⁴)	S _{xe} (in ³)	M _{al} (in-k)	V _{ay} (kip)	Jx1000 (in ⁴)	C _w (in ⁶)	X _o (in.)	r _o (in.)	β
800T125-68	0.0713	50	0.7298	2.4870	5.5464	1.3866	2.757	0.0699	0.310	5.0162	1.1717	35.081	4.220	1.2366	0.8573	-0.435	2.808	0.9760
800T125-97	0.1017	50	1.0294	3.5082	7.7011	1.9253	2.735	0.0960	0.305	7.7035	1.9267	57.684	10.888	3.5490	1.1714	-0.428	2.785	0.9763
800T150-33 ⁵	0.0346	33	0.3754	1.2794	3.0413	0.7603	2.846	0.0598	0.399	2.0401	0.4059	8.020	0.474	0.1498	0.7207	-0.594	2.935	0.9591
800T150-43	0.0451	33	0.4884	1.6645	3.9437	0.9859	2.842	0.0771	0.397	3.0154	0.6408	12.663	1.051	0.3311	0.9277	-0.591	2.929	0.9593
800T150-54	0.0566	50	0.6107	2.0812	4.9062	1.2265	2.834	0.0956	0.396	3.8243	0.8211	24.584	2.091	0.6521	1.1482	-0.588	2.922	0.9595
800T150-68	0.0713	50	0.7654	2.6085	6.1067	1.5267	2.825	0.1186	0.394	5.2939	1.2106	36.244	4.220	1.2970	1.4210	-0.585	2.911	0.9596
800T150-97	0.1017	50	1.0803	3.6815	8.4941	2.1235	2.804	0.1639	0.390	8.2968	2.0510	61.406	10.888	3.7243	1.9533	-0.578	2.889	0.9600
800T200-33 ⁵	0.0346	33	0.4100	1.3974	3.5901	0.8975	2.959	0.1346	0.573	2.2239	0.4329	8.554	0.474	0.1636	1.5707	-0.926	3.153	0.9138
800T200-43	0.0451	33	0.5335	1.8182	4.6572	1.1643	2.955	0.1742	0.571	3.4195	0.7276	14.377	1.051	0.3617	2.0267	-0.923	3.148	0.9141
800T200-54	0.0566	50	0.6673	2.2741	5.7990	1.4497	2.948	0.2166	0.570	4.3635	0.9411	28.177	2.091	0.7126	2.5152	-0.920	3.140	0.9142
800T200-68	0.0713	50	0.8367	2.8515	7.2272	1.8068	2.939	0.2697	0.568	6.1574	1.4247	42.656	4.220	1.4179	3.1235	-0.916	3.130	0.9143
800T200-97	0.1017	50	1.1820	4.0281	10.0802	2.5201	2.920	0.3754	0.564	9.2466	2.2133	66.265	10.888	4.0750	4.3249	-0.909	3.110	0.9146
1000T125-43 ⁵	0.0451	33	0.5561	1.8950	6.3630	1.2726	3.383	0.0473	0.292	4.8019	0.8052	15.910	0.836	0.3770	0.9382	-0.383	3.417	0.9874
1000T125-54	0.0566	50	0.6956	2.3706	7.9219	1.5844	3.375	0.0585	0.290	6.0989	1.0327	30.918	1.660	0.7428	1.1594	-0.381	3.409	0.9875
1000T125-68	0.0713	50	0.8724	2.9730	9.8704	1.9741	3.364	0.0724	0.288	8.4993	1.5308	45.831	3.345	1.4783	1.4319	-0.378	3.397	0.9876
1000T125-97	0.1017	50	1.2328	4.2014	13.7585	2.7517	3.341	0.0995	0.284	13.4455	2.6329	78.829	9.862	4.2503	1.9596	-0.371	3.373	0.9879
1000T150-43 ⁵	0.0451	33	0.5786	1.9719	6.9217	1.3843	3.459	0.0802	0.372	4.9997	0.8224	16.252	0.836	0.3923	1.5537	-0.518	3.517	0.9783
1000T150-54	0.0566	50	0.7239	2.4670	8.6214	1.7243	3.451	0.0995	0.371	6.3563	1.0559	31.613	1.660	0.7730	1.9242	-0.516	3.509	0.9784
1000T150-68	0.0713	50	0.9080	3.0945	10.7498	2.1498	3.441	0.1235	0.369	8.9098	1.5762	47.191	3.345	1.5387	2.3832	-0.512	3.498	0.9785
1000T150-97	0.1017	50	1.2837	4.3747	15.0040	3.0008	3.419	0.1706	0.365	14.3431	2.7782	83.179	9.862	4.4256	3.2814	-0.506	3.475	0.9788
1000T200-43 ⁵	0.0451	33	0.6237	2.1256	8.0391	1.6078	3.590	0.1826	0.541	5.3358	0.8461	16.720	0.836	0.4229	3.4109	-0.820	3.722	0.9514
1000T200-54	0.0566	50	0.7805	2.6599	10.0205	2.0041	3.583	0.2271	0.539	6.7929	1.0876	32.563	1.660	0.8335	4.2359	-0.818	3.715	0.9515
1000T200-68	0.0713	50	0.9793	3.3375	12.5061	2.5012	3.574	0.2829	0.537	9.6032	1.6383	49.051	3.345	1.6595	5.2648	-0.814	3.704	0.9517
1000T200-97	0.1017	50	1.3854	4.7213	17.4951	3.4990	3.554	0.3940	0.533	15.7308	2.9526	88.401	9.862	4.7762	7.3024	-0.807	3.683	0.9520
1200T125-54 ⁵	0.0566	50	0.8088	2.7564	12.7762	2.1294	3.974	0.0600	0.272	9.2473	1.2638	37.839	1.377	0.8637	1.7538	-0.337	3.998	0.9929
1200T125-68	0.0713	50	1.0150	3.4590	15.9401	2.6567	3.963	0.0742	0.270	13.0210	1.8900	56.585	2.770	1.7199	2.1669	-0.334	3.986	0.9930
1200T125-97	0.1017	50	1.4362	4.8946	22.2831	3.7138	3.939	0.1019	0.266	20.9847	3.3218	99.455	8.145	4.9515	2.9683	-0.328	3.962	0.9931
1200T150-54 ⁵	0.0566	50	0.8371	2.8528	13.7854	2.2976	4.058	0.1023	0.350	9.5967	1.2904	38.634	1.377	0.8939	2.9218	-0.459	4.099	0.9874
1200T150-68	0.0713	50	1.0506	3.5805	17.2083	2.8681	4.047	0.1270	0.348	13.5845	1.9419	58.141	2.770	1.7803	3.6205	-0.456	4.088	0.9875
1200T150-97	0.1017	50	1.4871	5.0679	24.0828	4.0138	4.024	0.1755	0.344	22.2554	3.4914	104.532	8.145	5.1268	4.9901	-0.450	4.064	0.9877
1200T200-54 ⁵	0.0566	50	0.8937	3.0457	15.8038	2.6340	4.205	0.2350	0.513	10.1866	1.3267	39.722	1.377	0.9543	6.4649	-0.737	4.300	0.9706
1200T200-68	0.0713	50	1.1219	3.8235	19.7447	3.2908	4.195	0.2927	0.511	14.5286	2.0125	60.253	2.770	1.9011	8.0394	-0.734	4.289	0.9708
1200T200-97	0.1017	50	1.5888	5.4145	27.6822	4.6137	4.174	0.4079	0.507	24.1808	3.6902	110.486	8.145	5.4775	11.1634	-0.727	4.267	0.9710

¹Use the effective moment of inertia for deflection calculation.

²The effect of cold-work of forming on strength was conservatively not incorporated.

³See technical manual for product identification information. Calculations were performed using dimensions specified in the client's technical manual dated 7/29/2025.

⁴Flange width-to-thickness ratio exceeds 60. Allowable design values for these members are outside the scope and may be determined in accordance with AISI S100.

⁵Web height-to-thickness ratio, h/t , exceeds 200 but is less than 260. Bearing stiffeners are required in accordance with AISI S100. No holes/punchouts are permitted in the web at these locations.

SYMBOLS

I_x = Moment of inertia about x-axis
 S_x = Section modulus about x-axis
 r_x = Radius of gyration about x-axis
 I_y = Moment of inertia about y-axis
 r_y = Radius of gyration about y-axis

M_{ai} = Allowable bending moment about x-axis based on local buckling
 V_{ay} = Allowable shear

J = St. Venant torsion constant
 C_w = Torsional warping constant
 x_o = Distance from shear center to the centroid along the principal X-axis
 r_o = Polar radius of gyration about shear center
 β = Torsional flexural constant